

David W Miller

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Automation Engineer

Automation Engineer with 5+ years building production pipelines, CI/CD systems, and developer tooling at scale. Deep expertise in Python scripting, AWS automation, and infrastructure-as-code. Designed and maintained build systems for high-impact products including Fortnite, enabling distributed engineering teams to ship reliably to millions of users. Proven track record identifying workflow bottlenecks, implementing automated solutions, and building reusable tools that reduce manual effort and accelerate delivery. Strong with Jenkins pipelines, Docker, REST APIs, and cross-team collaboration. Excels at translating complex technical problems into practical automation that just works.

TECHNICAL SKILLS

CI/CD & Build Tools

Jenkins (pipeline development, automation, administration) • Version control (Perforce Helix Core/Swarm, GitHub) • GitHub Actions • Artifact management

Infrastructure & Cloud AWS

AWS (CLI, EC2, S3, IAM) • Docker • Distributed computing infrastructure • New Relic • Sentry • Infrastructure automation

Programming & Scripting

Python (2.x-3.x: PyQt/PySide, REST APIs, automation tooling) • Groovy • C++ • C# • Bash/Shell scripting

Production Systems

Render farm orchestration (AWS Deadline) • Container workflows • MongoDB • Asset pipeline architecture • Deployment automation

EXPERIENCE

Technical Director | Mar 2021 - Mar 2026

HALON Entertainment

Engineered and maintained scalable build infrastructure and deployment automation supporting 11 distributed production environments. Developed CI/CD tooling, pipeline orchestration systems, and infrastructure automation that enabled high-throughput software delivery and asset management at scale. Collaborated cross-functionally with development teams to evaluate existing pipelines, implement shared solutions, and establish DevOps best practices across multiple projects.

- Automated Unreal Engine build triggering and packaging workflows using Jenkins and Groovy scripting, integrating with Perforce/UnrealGameSync for version control synchronization and automated deployment.
- Implemented production monitoring and observability stack (New Relic, Sentry), establishing performance baselines, error tracking, and alerting workflows that reduced incident response time and improved system reliability.
- Deployed and maintained AWS Deadline distributed render farm infrastructure, automating job scheduling, resource allocation, and failure recovery to support production at scale.
- Designed and implemented an in-house branch of the open-source AYON Unreal Engine Python plugin, a backend integration system managing distributed asset workflows and pipeline orchestration across multiple projects and teams.
- Deployed and administered AWS EC2 Unreal Engine Multi-User session environment, implementing automated resource allocation and failure recovery to support live capture production workloads.
- Resolved 100+ engineering tickets spanning infrastructure automation, API integrations, system migrations, and deployment bug fixes, improving pipeline efficiency and reducing deployment friction for development teams.
- Performed code reviews for Python-based render pipeline serving all production environments, ensuring stable integrations across build systems, asset publishing, and deployment infrastructure.
- Provided DevOps engineering support via Jira ticketing system, troubleshooting environment configuration issues, debugging software integration issues, and resolving tooling failures across Maya and Unreal Engine build systems.

Technical Artist | Oct 2017 - Mar 2021

HALON Entertainment

- Provided technical art support for numerous virtual production shoots, troubleshooting real-time rendering issues, camera tracking integration, and LED wall calibration during live capture sessions.
- Authored standardized PBR materials and pipeline tools used across the studio on all projects including custom layering controls and feature sets, expanding artist capabilities and saving time.
- Built Unreal Engine to Maya MetaHuman export and import tools in Python, including blendshape transfer and skin weight preservation for supporting project animation workflow.
- Implemented crowd simulation systems and procedural animation rigs for large-scale background character sequences in Unreal Engine productions.

Engine Artist & Game Dev | Feb 2017 - Aug 2017

MatterVR

- Developed mobile VR game Wonders of the World for GearVR using early Unreal Engine 4.
- Performance profiled and optimized project assets for VR rendering, diagnosing frame-rate bottlenecks and memory issues to maintain consistent 60fps targets across GearVR hardware constraints.
- Built multiplayer tabletop minigames in BP & C++, featuring historical games Knucklebones, Nine Men's Morris and Senet.
- Developed a custom data-driven, branching Quest system, managing integration and game development testing.

JSC Distance Learning Instructional Designer | July 2014 - Oct 2016

NASA Johnson Space Center, GHG Corporation

- Delivered +200 interactive online lessons to the Flight Operations Directorate software familiarization and classroom-based instruction, providing refresher briefings and training courses to flight controllers, instructors and astronauts.
- Developed customized training overviews for ISS laptop software by meeting with stakeholders and developers and hosting planning sessions.
- SharePoint developer & administrator, supporting instructors and division pages across the Training Standards & Integration Production Group.
- Prototyped a fully-functional ISS Environment in Unreal Engine 4 for Interactive Emergency Training.

JSC Distance Learning Training Standards Intern | Sep 2011 - Jul 2014

NASA Johnson Space Center, United Space Alliance/LZ Technology

- Built 3 ISS Emergency Response online training courses using Articulate Storyline for Fire, Depress or Toxic Atmosphere scenarios using training footage from station mockups.
- Camera Operator for Video Lab capture of Flight Controller and Astronaut training footage and briefings.

EDUCATION

M.A. Digital Media Studies / University of Houston - Clear Lake | Dec 2016

Compositing, Illustration, 3D Modelling & Animation

B.A. Communications / University of Houston - Clear Lake | May 2011

TV Production Studio, School Newspaper & Magazine